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JSS SCIENCE AND TECHNOLOGY UNIVERSITY



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JSS TECHNICAL INSTITUTIONS CAMPUS
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Event 2 Report on

“Smart Library System”

Subject Name: Principles of Electrical Engineering

Subject Code: 24CB140

CO5: Describe the elementary methods for measurement of electrical quantities in DC and AC systems and the concept of sensors, transducers, domestic electrical installation and safety

Submitted by

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ABSTRACT

This project presents an RFID-based Library Management System designed to simplify and automate book issue and return operations. In this system, RFID tags are attached to books and RFID cards are assigned to users. When a book is scanned for the first time, the system identifies the transaction as book taken and updates the status accordingly. When the same book is scanned again, the system automatically recognizes it as book returned and updates the record. All transaction details such as book ID, title, author, user ID, date, time, and status are uploaded in real time to a Google Spreadsheet using a Wi-Fi enabled microcontroller.

The proposed system reduces manual work and eliminates the need for separate issue and return procedures. It improves accuracy, saves time, and maintains real-time digital records of library transactions. The system is cost-effective, easy to implement, and suitable for schools, colleges, and small libraries. This project demonstrates the effective use of RFID and IoT technologies for efficient library management.

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CHAPTER 1

1. Introduction:

Libraries play an important role in schools and colleges by providing students with access to books and learning resources. In many libraries, book issue and return work is still done manually or using basic computer systems. These traditional methods require a lot of time and effort and are often affected by human errors. Maintaining registers, checking book availability, and updating records manually can be difficult, especially when the number of books and users is large. Therefore, there is a strong need for an automated system that can make library management faster, easier, and more accurate.

Radio Frequency Identification (RFID) technology provides a simple and effective solution to this problem. RFID uses radio waves to identify objects uniquely. In this system, RFID tags are attached to library books, and RFID cards are issued to users. When a book or card is scanned using an RFID reader, the information stored in the tag is read automatically without physical contact. Unlike barcode systems, RFID does not require direct line-of-sight scanning, which makes the process faster and more reliable.

The RFID-based Library Management System developed in this project automates the book issue and return process. When a book is scanned for the first time, the system displays the message “Book Taken”, indicating that the book has been issued. When the same book is scanned again, the system automatically displays “Book Returned”, showing that the book has been returned to the library. This automatic status change removes confusion and reduces the need for manual verification by the librarian.

To store and manage data efficiently, the system uses a Wi-Fi enabled microcontroller to upload all transaction details to a Google Spreadsheet. The stored data includes book ID, book name, author, user details, date, time, and transaction status. This allows library staff to view and manage records in real time from any location. Cloud-based storage also ensures data safety and easy backup.

Overall, this system reduces paperwork, saves time, and improves accuracy in library operations. It is cost-effective, easy to use, and suitable for schools, colleges, and small libraries. By using RFID and IoT technologies, the project helps in transforming traditional libraries into smart and automated systems.

2. Motivation:

Libraries are important learning centers, but many still use traditional methods to manage book issue and return activities. Manual record-keeping and basic systems require more time, effort, and staff involvement. These methods often lead to errors such as incorrect entries, missing records, and difficulty in tracking book availability. During peak hours, handling multiple transactions becomes challenging and inefficient.

With the growth of digital technologies, there is a strong need for an automated and reliable library management system. RFID and IoT technologies offer a practical solution by enabling fast, contactless identification of books and automatic data storage. This project is motivated by the idea of simplifying the issue and return process using a smart scanning mechanism that clearly displays “Book Taken” and “Book Returned”, while maintaining real-time digital records.

- Traditional library systems rely on manual record-keeping, which is time-consuming and prone to human errors.
- Maintaining issue and return registers becomes difficult when the number of books and users increases.
- There is confusion and delay in identifying whether a book is issued or returned in busy library environments.
- RFID technology enables fast, contactless, and accurate identification of books and users.
- The concept of showing “**Book Taken**” on the first scan and “**Book Returned**” on the next scan simplifies the transaction process.
- Cloud-based data storage allows real-time access, easy monitoring, and secure backup of library records.
- The project aims to reduce paperwork, save time, and improve overall efficiency in library management.

3. OBJECTIVES:

- To design and implement an RFID-based system for automating the book issue and return process.
- To develop a mechanism where the first scan displays “Book Taken” and the next scan displays “Book Returned.”
- To store and manage library transaction data digitally for easy access and monitoring.

CHAPTER 2

METHODOLOGY:

The methodology of the RFID-based Library Management System explains how the system works step by step. Initially, RFID tags are attached to all books, and RFID cards are issued to users. Each RFID tag contains a unique identification number that represents a particular book. An RFID reader is used to scan these tags whenever a book is issued or returned.

When a book is scanned, the RFID reader reads the tag ID and sends it to the Wi-Fi enabled microcontroller. The microcontroller checks the previous status of the book stored in the database. If the book is scanned for the first time, the system marks it as “Book Taken.” When the same book is scanned again, the system automatically updates the status to “Book Returned.” This logic eliminates the need for separate issue and return operations.

After determining the status, the microcontroller sends the transaction details such as book ID, status, date, and time to a cloud-based Google Spreadsheet using an internet connection. The data is stored in real time and can be accessed by authorized users from anywhere. A display unit (such as an LCD or serial monitor) shows the current book status to the user.

This methodology ensures faster transactions, reduces human errors, and maintains accurate digital records of library activities.

Block diagram (in points):

- RFID Tag: Stores a unique identification number for each book.
- RFID Reader: Reads the book’s RFID tag when it is scanned.
- Microcontroller: Processes the tag data and decides whether the book is *taken* or *returned*.
- Display Unit: Shows the status message (*Book Taken* / *Book Returned*).
- Wi-Fi Module: Sends transaction data to the cloud.
- Cloud Platform (Google Sheets): Stores and manages library records digitally.
- Power Supply: Provides required power to all system components.

DEVICE USED:

1. ESP8266:

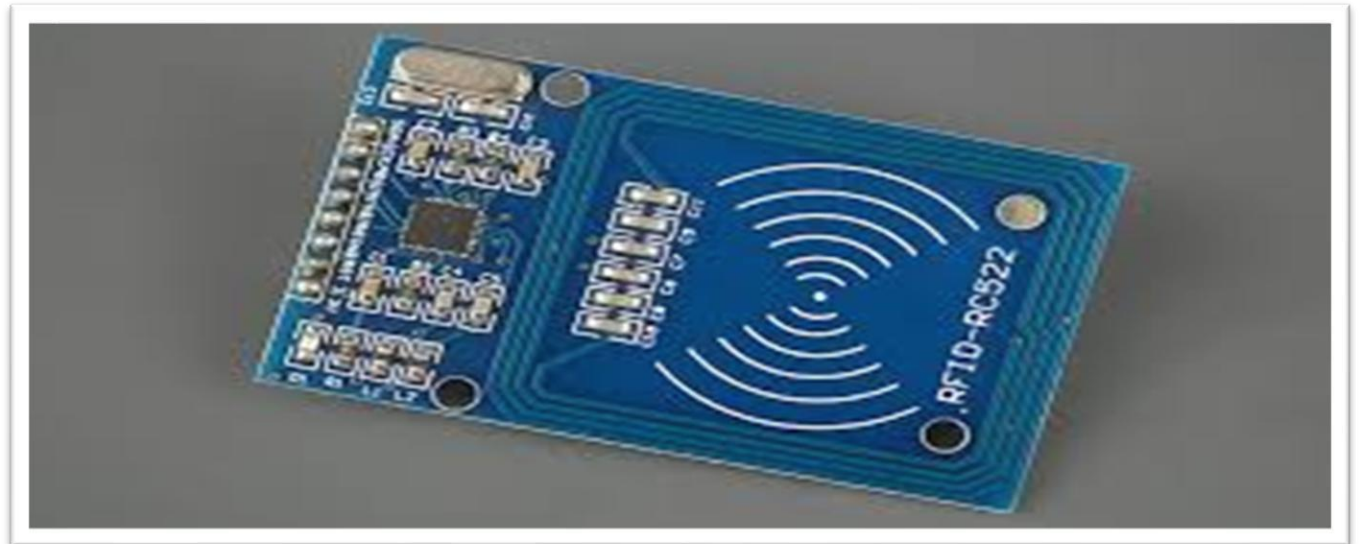
The ESP8266 is a Wi-Fi enabled microcontroller used as the main controller in our project. It processes the data received from the RFID reader and connects to the internet. Using its built-in Wi-Fi module, it sends and updates book information to an online database or Google Sheets in real time. The ESP8266 makes the system smart, fast, and wireless.



2. RFID READER (MFRC522):

The MFRC522 is an RFID reader module used to read data from RFID cards and tags. It operates at 13.56 MHz and communicates with the ESP8266 using the SPI protocol. When a card or tag is scanned, the

MFRC522 reads its unique ID and sends it to the controller for verification. This module plays a key role in identifying books and users in the library system.



3. RFID TAGS:

RFID cards and tags are used to store unique identification numbers. In this project, each book or user is assigned a specific RFID tag. When the tag is brought near the RFID reader, it is detected without physical contact. This helps in quick book issuing and returning, reducing manual work and errors.



4. POWER SUPPLY:

In this project, a USB cable is used as the main power supply for the system. The USB provides a stable 5V power to the ESP8266, which is internally regulated to the required operating voltage. Apart from supplying power, the USB connection is also used to upload the program into the ESP8266. Using USB power makes the system easy to use, safe, and suitable for testing and demonstration purposes.

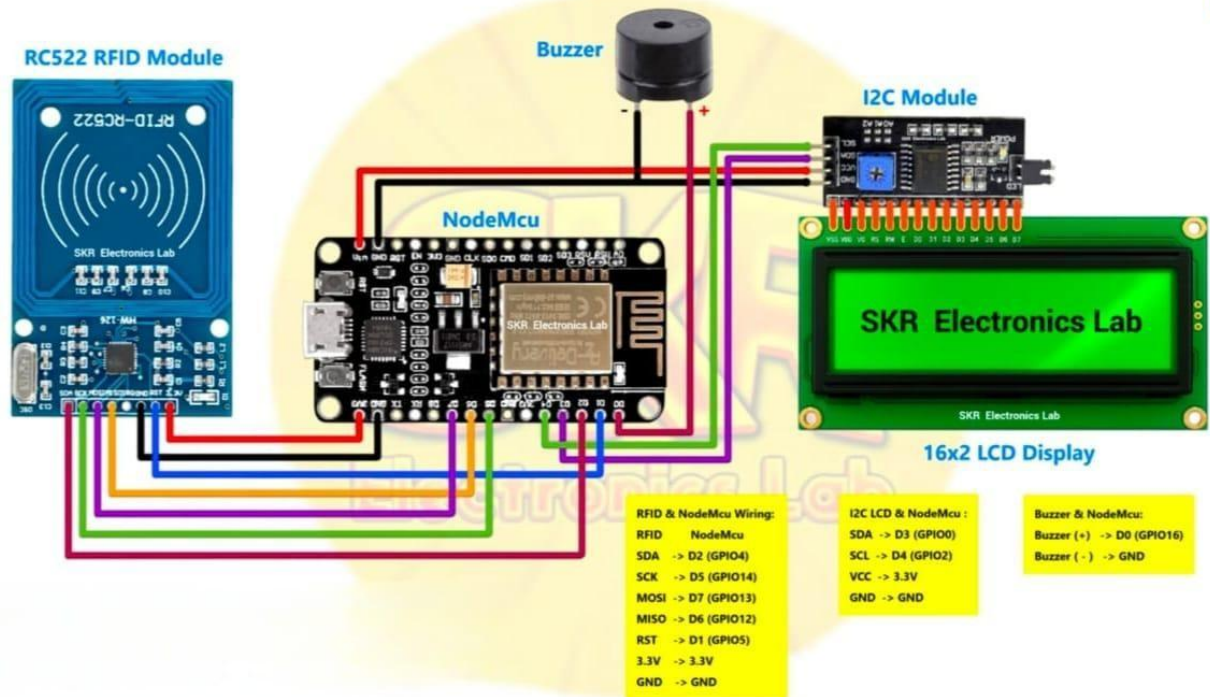


5. BUZZER:

In this project, a buzzer is used as an alert and indication device. It provides an audio signal when an RFID card or tag is successfully scanned. The buzzer helps the user know whether the book is issued or returned correctly. It improves user interaction and gives instant feedback during system operation.



Circuit diagram:



CHAPTER 3

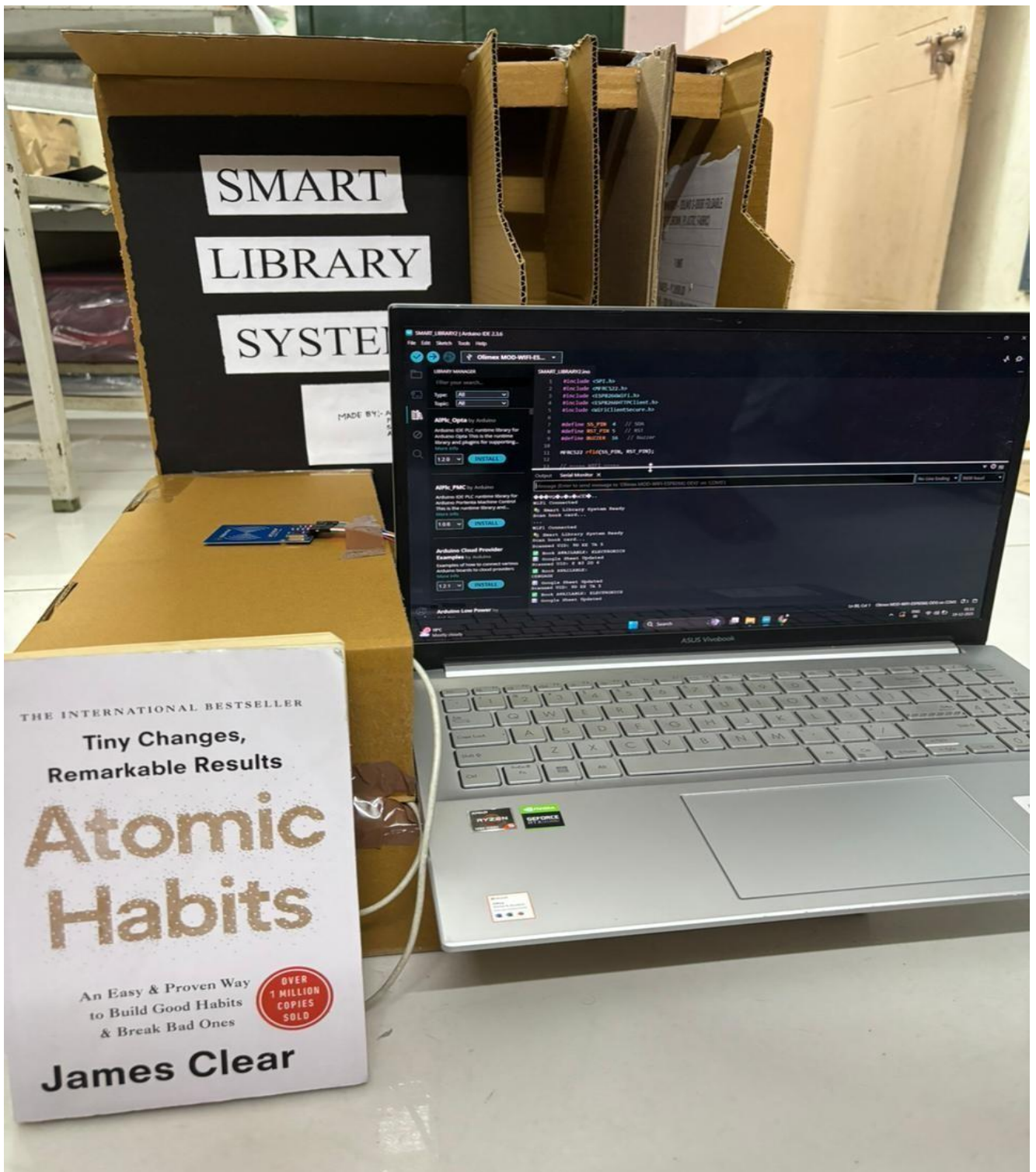
1. RESULT AND DISCUSSION:

- The system successfully automates book issue and return using RFID technology.
- On the first scan of a book, it displays “Book Taken”, indicating the book is issued.

- On scanning the same book again, it displays “Book Returned”, updating the status automatically.
- The display unit provides clear, instant feedback to the user and librarian.
- All transaction details (book ID, name, author, date, time, status) are uploaded in real time to Google Sheets.
- Library staff can access and monitor records from anywhere via the cloud platform.
- The system reduces manual work, paperwork, and human errors.
- Testing shows accurate identification of multiple books and users without errors.
- Real-time updates and automatic status tracking improve efficiency in library management.
- The project demonstrates a reliable, cost-effective, and user-friendly solution for modern libraries

WORKING MODEL:





CHAPTER 4

1. CONCLUSION:

The RFID-based Library Management System successfully automates the process of issuing and returning books. By using RFID tags, readers, and a microcontroller, the system can identify books and update their status as “Book Taken” or “Book Returned” automatically. All transaction details are stored digitally in real time on a cloud platform, reducing paperwork and manual errors. The system is fast, accurate, and user-friendly, making it highly suitable for schools, colleges, and small to medium-sized libraries. Overall, the project demonstrates the effective integration of RFID and IoT technologies to make library operations efficient and modern.

2. FUTURE SCOPE:

- Integration with library management software for advanced features like automated notifications for due dates.
- Use of mobile apps to allow students and staff to check book availability remotely.
- Implementation of multi-user simultaneous scanning to handle busy library environments.
- Addition of security features such as alerts for unauthorized book removal.
- Expansion to large-scale libraries with multiple sections and automated inventory management.
- Incorporation of data analytics to track book usage patterns and improve library services.

REFERENCE:

1. How to setup and program nodeMCU ESP8266 (Youtube -Binary updates).
2. ESP8266 explanation (Mr DIY)
3. What is RFID? How RFID works? RFID Explained in Detail (AAE).
4. Google Sheet Home Automation NodeMCU 8266 || esp8266 read google sheet || esp8266 google spreadsheet(Teach me something).
5. <https://www.espressif.com/en/products/socs/esp8266> all information about the ESP8266.

PROJECT CODE:

GOOGLE DRIVE:

https://docs.google.com/document/d/17MCEvRguAk6BaBB81L5DTSYliU1Yyfl_DQ1IChFeono/edit?usp=drive_link

GITHUB:

<https://github.com/anurag-rawat-hub/smart-library.git>

PROJECT CODE:

```
#include <SPI.h>
```

```

#include <MFRC522.h>

#include <ESP8266WiFi.h>

#include <ESP8266HTTPClient.h>

#include <WiFiClientSecure.h>


#define SS_PIN 4

#define RST_PIN 5

#define BUZZER 16


MFRC522 rfid(SS_PIN, RST_PIN);


// ===== WIFI ===== const char*
ssid = "iqoo"; const char* password
= "pankaj90";


// ===== GOOGLE SCRIPT URL =====
String googleURL =
"https://script.google.com/macros/s/AKfycbyu5YWkWlxlz83FGoUBu57yGXiK5tKtTqp2MJ2BiKaSONNLXzYrRNZL_IaUWC7AQfaxQ/exec";


// ===== BOOK DATABASE =====

byte book1[4] = {0x9D, 0xEE, 0x7A, 0x05}; String
book1Name = "ELECTRONICS";

bool book1Issued = false;


byte book2[4] = {0x0E, 0xB3, 0x2D, 0x06}; String
book2Name = "CENGAGE";

bool book2Issued = false;

void setup() {
  Serial.begin(9600);

```

```

SPI.begin();
rfid.PCD_Init();

pinMode(BUZZER, OUTPUT);
digitalWrite(BUZZER, LOW);

Serial.println("=====");
Serial.println("  SMART LIBRARY SYSTEM (RFID)  ");
Serial.println("=====");

Serial.print("[■] Connecting to WiFi");
WiFi.begin(ssid, password); while
(WiFi.status() != WL_CONNECTED) {
    delay(300);
    Serial.print(".");
}

Serial.println("\n[■■] WiFi Connected");
Serial.println("[■■■] System Ready");
Serial.println("[■] Waiting for Scan...");
}

void loop() {

    if (!rfid.PICC_IsNewCardPresent()) return;
if (!rfid.PICC_ReadCardSerial()) return;
Serial.println("\n[■■] RFID Detected");

    // ===== READ UID =====

```

```

String uidStr = ""; for (byte i = 0; i < 4;
i++) { uidStr += String(rfid.uid.uidByte[i],
HEX);
}
uidStr.toUpperCase();

// ===== CHECK BOOK =====
if (compareUID(rfid.uid.uidByte, book1)) {
handleBook(book1Name, book1Issued, uidStr);
}
else if (compareUID(rfid.uid.uidByte, book2)) {
handleBook(book2Name, book2Issued, uidStr);
}
else {
Serial.println("[ERROR] Unknown Book");
rfid.PICC_HaltA(); return;
}

Serial.println("[■] Waiting for Scan...");
rfid.PICC_HaltA(); delay(1500);
}

// ===== FUNCTIONS =====

void handleBook(String bookName, bool &issuedStatus, String uid) {
if (!issuedStatus) {
Serial.print("[TAKEN] Book Issued: ");
Serial.println(bookName); issuedStatus = true;
beepShort(); sendToGoogleSheet(uid, bookName +
" - TAKEN");
}
}

```

```

    }
else {
    Serial.print("[RECEIVED] Book Returned: ");
    Serial.println(bookName);    issuedStatus = false;
    beepLong();    sendToGoogleSheet(uid, bookName + " -
    RECEIVED");
    }
}

```

```

bool compareUID(byte *a, byte *b) {
    for (byte i = 0; i < 4; i++) {
        if (a[i] != b[i]) return false;
    }
    return true;
}

```

```

void beepShort() {
    digitalWrite(BUZZER, HIGH);
    delay(200);    digitalWrite(BUZZER,
    LOW);
}

```

```

void beepLong() {
    digitalWrite(BUZZER, HIGH);
    delay(800);    digitalWrite(BUZZER,
    LOW);
}

```

```

void sendToGoogleSheet(String uid, String bookStatus) {

```



```
    if (WiFi.status() != WL_CONNECTED) {  
Serial.println("[ERROR] WiFi Disconnected");  
        return;  
    }  
  
    WiFiClientSecure client;  
    client.setInsecure();  
  
    HTTPClient https;  
    String url = googleURL + "?uid=" + uid + "&book=" + bookStatus;  
  
    https.begin(client, url);  
    https.GET();  https.end();  
  
    Serial.println("[✓] Google Sheet Updated");  
}
```